

MBA Fastrack 2025 (CAT + OMETs)

QUANTITATIVE APTITUDE

DPP: 3

3D Geometry + Polygon

- Q1** A solid metallic sphere of radius 3 cm is melted and recast into a solid cylinder of radius 1 cm. What is the height (in cm) of the cylinder?
- Q2** A metallic cone of height 24 cm and radius 6 cm is melted and recast into small spherical balls, each of diameter 2 cm. How many such spherical balls can be formed ?
- Q3** A hollow spherical shell has external radius 9 cm and internal radius 5 cm. It is melted and recast into a solid cone of height 8 cm. Find the radius of the base of the cone (in cm)
 (A) $\sqrt{307}$ (B) $\sqrt{310}$
 (C) $\sqrt{302}$ (D) $\sqrt{304}$
- Q4** A right circular cone has a base radius of 6 cm and a height of 8 cm. What is the maximum volume (in cm^3) of a sphere that can be inscribed in this cone?
 (A) 36π (B) 72π
 (C) 108π (D) 144π
- Q5** A solid cuboid has dimensions of 10 cm, 8 cm, and 6 cm. What is the maximum volume of a sphere (in cm^3) that can be placed inside this cuboid?
 (A) 8π (B) 18π
 (C) 27π (D) 36π
- Q6** A metallic cube has a side length of 10 cm. What is the maximum volume (in cm^3) of a sphere that can be carved out from this cube ?
 (A) $\frac{250\pi}{6}$
 (B) $\frac{500\pi}{3}$
 (C) $\frac{500\pi}{6}$
 (D) $\frac{250\pi}{3}$
- Q7** A sphere has a radius of 10 cm. What is the maximum volume (in cm^3) of a cube that can be inscribed in this sphere?
 (A) $\frac{8000\sqrt{3}}{3}$ (B) $\frac{4000\sqrt{3}}{9}$
 (C) $\frac{8000\sqrt{3}}{9}$ (D) $\frac{7000\sqrt{3}}{2}$
- Q8** A right circular cone has a height of 18 cm and a base radius of 6 cm. It is cut by a plane parallel to its base. The cut is made at a height of 6 cm from the base. What is the ratio of the volume of the smaller cone formed at the top to the volume of the frustum formed at the bottom?
 (A) $\frac{8}{11}$ (B) $\frac{8}{19}$
 (C) $\frac{4}{19}$ (D) $\frac{8}{15}$
- Q9** A right circular cone is cut by a plane parallel to its base. The volume of the smaller cone formed at the top is $\frac{1}{64}$ of the volume of the original cone. If the height of the frustum (the lower part) is 21 cm, what was the original height (in cm) of the cone?
- Q10** A right circular cone is cut by a plane parallel to its base. The cut is made such that the curved surface area of the frustum (the lower part) is $\frac{7}{9}$ times the curved surface area of the original cone. If the height of the smaller cone formed at the top is 4 cm, what was the original height (in cm) of the cone?
 (A) $5\sqrt{2}$ (B) $6\sqrt{2}$
 (C) $9\sqrt{2}$ (D) $6\sqrt{3}$



Q11 The top of a conical vessel has a circumference of 352 m. Water flows in at a rate of $60 \text{ m}^3/\text{sec}$. Find the approximate time (in minutes) taken for the vessel to be 75% full if its depth is 60 m.
 (A) 41 (B) 65
 (C) 53 (D) None of these

Q12 Find the total surface area in sq. cm. of a hollow cylinder, open at both ends, whose external diameter is 18 cm, thickness is 3 cm and the height is 40 cm.
 (A) 1230π (B) 1650π
 (C) 1080π (D) 1290π

Q13 The sum of the radius of the base of a solid cylinder and the height of the cylinder is 15 cm. If the total surface area of the cylinder is 660 cm^2 , then find the volume of the cylinder (in cubic cm)
 (A) 1458 (B) 1232
 (C) 1396 (D) None of these

Q14 The interior of a building is in the form of a cylinder of diameter $12\sqrt{3} \text{ m}$, surmounted by a cone of vertical angle 60 degrees. Find the volume of the building in meter cubic if the overall height of the building is 30 m.
 (A) 1944π (B) 1856π
 (C) 2105π (D) 2446π

Q15 The radii of a cylinder and a cone are equal and their heights are also equal. If the ratio of curved surface area of the cone to that of the cylinder is 17 : 16, then find the ratio of the radius to height for the cylinder
 (A) 15 : 8 (B) 13 : 9
 (C) 17 : 7 (D) None of these

Q16 If the length of the smaller diagonal of a regular hexagon is given as 14 cm. Find the area of the hexagon.
 (A) $198\sqrt{3} \text{ cm}^2$ (B) $98\sqrt{3} \text{ cm}^2$
 (C) $196\sqrt{3} \text{ cm}^2$ (D) $99\sqrt{3} \text{ cm}^2$

Q17 Find the value of 'x' given that the difference between an exterior angle of $(x + 3)$ sided regular polygon and an exterior angle of $(x - 2)$ sided polygon is 12 degrees.
 (A) 12
 (B) -13
 (C) 11
 (D) 23

Q18 In the figure given below, a circle is inscribed inside a regular hexagon and the hexagon itself is inscribed in another circle. Find the ratio of the circumradius of the regular hexagon to its inradius



(A) $\sqrt{3} : 2$ (B) $1 : \sqrt{6}$
 (C) $2 : \sqrt{3}$ (D) $\sqrt{3} : 1$

Q19 Each of the interior angles of a regular polygon containing 'n' sides is equal to 170. Find the value of 'n'

Q20 ABCDEF is a regular hexagon such that AD = 4 cm, then find the area of quadrilateral ABDF (in cm^2).
 (A) $2\sqrt{3}$ (B) $4\sqrt{3}$
 (C) $6\sqrt{3}$ (D) $8\sqrt{3}$

Q21 If the sum of the interior angles of a regular polygon is eight times the sum of its exterior angles, how many diagonals does the polygon have?

Q22 Given that there are two regular polygons with the ratio between their number of sides as 2 : 3 and the ratio of measures of each interior angle is 3 : 4. What is the difference between the number of sides of the two polygons?



- Q23** PQRSTU is a regular hexagon. M, N, O, and L are the midpoints of PQ, QR, TU, and UP, respectively. What is the area of the quadrilateral MNOL in square centimeters if the area of the hexagon is 360 cm^2 ?
- Q24** ABCDEF is a regular hexagon such that the diagonals EC and AD intersect at G. If $AG = 3 \text{ cm}$, then find the side length of the hexagon (in cm).
- Q25** If the length of a side of a regular hexagon PQRSTU is 6 cm and V is the midpoint of ST, find the length of PV.
- (A) $\sqrt{117} \text{ cm}$ (B) $\sqrt{113} \text{ cm}$
(C) $\sqrt{111} \text{ cm}$ (D) $\sqrt{108} \text{ cm}$
- Q26** A solid right circular cone has a height of 35 cm. The top of the cone was sawed off at a height of 21 cm from the base. What is the volume of the resultant frustum of cone in cubic centimeters if the difference between the volumes of the two resultant pieces is 7194 cubic centimeters?
- Q27** A rope is wrapped around a cylindrical pole starting from the bottom at a specific angle, as shown in the diagram. The top end of the rope is directly above the bottom end. If the diameter and height of the pole are 7 cm and 24 m respectively, and the rope is wrapped around the pole 20 times, what is the total length of the rope used in centimeters?
- 
- (A) 2440 (B) 2560
(C) 2220 (D) 1920
- Q28** If water flows at a rate of 400 meters per hour through a cylindrical pipe with a diameter of 10.5 millimeters, how many seconds will it take to fill a tank that is shaped like a right pyramid

with a regular hexagonal base, where the height of the pyramid is $44\sqrt{3} \text{ cm}$ and the side length of the hexagonal base is 17.5 cm? (Use $\pi = 22/7$)

- Q29** Three solid cuboids made of three different materials each have square cross-sections with an area of 64 square centimeters. Each cuboid is melted and recast into identical smaller solid cubes made from a single material type. Assuming no metal wastage in the process, and knowing that each small cube has a total surface area of 384 square centimeters, which of the following triplets could represent the edge lengths (in centimeters) of the original cuboids?
- (A) 184, 168, 228 (B) 272, 228, 264
(C) 336, 144, 192 (D) 256, 448, 484
- Q30** A cuboid of maximum possible volume is carved out from a sphere of radius $7\sqrt{3} \text{ cm}$. Find the volume of the remaining sphere.
- (A) $5880\sqrt{3} \text{ cm}^3$
(B) $196(11\sqrt{3} - 7) \text{ cm}^3$
(C) $2940(\sqrt{3}) \text{ cm}^3$
(D) $392(11\sqrt{3} - 7) \text{ cm}^3$



Answer Key

Q1 36
Q2 216
Q3 C
Q4 A
Q5 D
Q6 B
Q7 C
Q8 B
Q9 28
Q10 B
Q11 A
Q12 D
Q13 B
Q14 A
Q15 A

Q16 B
Q17 A
Q18 C
Q19 36
Q20 B
Q21 135
Q22 2
Q23 135
Q24 2
Q25 A
Q26 7722
Q27 A
Q28 2100
Q29 C
Q30 D

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Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

Explanation:

When a solid is melted and recast, its volume remains constant.

$$\text{Volume of sphere} = \frac{4}{3}\pi R^3$$

$$\text{Volume of cylinder} = \pi r^2 h$$

Given, radius of sphere $R = 3$ cm.

Radius of cylinder $r = 1$ cm.

Let the height of the cylinder be h .

Equating the volumes

$$\frac{4}{3}\pi(3)^3 = \pi(1)^2 h$$

$$\frac{4}{3}\pi(27) = \pi(1)h$$

$$4 \times 9\pi = \pi h$$

$$36\pi = \pi h$$

$$h = 36 \text{ cm}$$

The height of the cylinder is 36 cm.

Video Solution:



Q2 Text Solution:

$$\text{Volume of cone} = \frac{1}{3}\pi r^2 h$$

$$\text{Volume of sphere} = \frac{4}{3}\pi R^3$$

Given, height of cone = 24 cm, radius of cone = 6 cm.

Diameter of each spherical ball = 2 cm, so radius of each spherical ball = 1 cm.

$$\begin{aligned} \text{Volume of the cone} &= \frac{1}{3}\pi(6)^2(24) \\ &= \frac{1}{3}\pi(36)(24) = 1\pi(12)(24) = 288\pi \text{ cm}^3 \end{aligned}$$

Volume of one spherical ball =

$$\frac{4}{3}\pi(1)^3 = \frac{4}{3}\pi \text{ cm}^3$$

Number of spherical balls formed =

Volume of cone / Volume of one spherical ball

$$= \frac{288\pi}{\frac{4}{3}\pi} = 288 \times \frac{3}{4} = 216$$

216 spherical balls can be formed.

Video Solution:



Q3 Text Solution:

Volume of hollow spherical shell =

$$\frac{4}{3}\pi (R^3 - r^3)$$

Volume of solid cone =

$$\frac{1}{3}\pi \times r^2 \text{ cone} \times h_{\text{cone}}$$

Given, external radius of shell $R_{\text{external}} = 9$ cm.

Internal radius of shell $R_{\text{internal}} = 5$ cm.

Height of cone $h_{\text{cone}} = 8$ cm.

Let the radius of the base of the cone be r_{cone} .

Volume of the hollow spherical shell =

$$\begin{aligned} \frac{4}{3}\pi(9^3 - 5^3) &= \frac{4}{3}\pi(729 - 125) \\ &= \frac{4}{3}\pi(604) \text{ cm}^3 \end{aligned}$$

Volume of the solid cone =

$$\frac{1}{3}\pi(r_{\text{cone}})^2 (8) \text{ cm}^3$$

ATP,

$$\frac{4}{3}\pi(604) = \frac{1}{3}\pi(r_{\text{cone}})^2 (8)$$

$$4 \times 604 = (r_{\text{cone}})^2 (8)$$

$$(r_{\text{cone}})^2 = \frac{2416}{8}$$

$$r_{\text{cone}} = \sqrt{302}$$

The radius of the base of the cone is $\sqrt{302}$ cm

Video Solution:



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**Q4 Text Solution:**

Let the radius of the cone be R and its height be H . The slant height L of the cone can be found using the Pythagorean theorem:

$$L = \sqrt{R^2 + H^2}$$

Given $R=6$ cm and $H=8$ cm:

$L =$

$$\sqrt{6^2 + 8^2} = \sqrt{36 + 64} = 10 = 10 \text{ cm..}$$

Let r be the radius of the largest sphere that can be inscribed in the cone.

The radius r of the largest sphere that can be inscribed in a cone is given by the formula:

$$r = RH/R+L$$

$$r = \frac{6 \times 8}{6+10} = \frac{48}{16} = 3 \text{ cm}$$

Calculate the maximum volume of the sphere:

$$\text{Volume of sphere} = \frac{4}{3}\pi r^3$$

$$\text{Volume} = \frac{4}{3}\pi \times 3^3$$

$$\text{Volume} = 36\pi \text{ cm}^3.$$

Video Solution:**Q5 Text Solution:**

For a sphere to be inscribed (or placed inside with maximum volume) within a cuboid, its diameter must be equal to the shortest dimension of the cuboid. If the diameter were larger than the shortest dimension, the sphere would not fit.

dimensions of the cuboid are:

Length (L) = 10 cm, Breadth (B) = 8 cm, Height (H) = 6 cm

shortest dimension of the cuboid is 6 cm.

Therefore, the diameter of the largest inscribed sphere = 6 cm.

The radius r of the largest inscribed sphere =

$$\frac{\text{Diameter}}{2} = \frac{6}{2} = 3 \text{ cm.}$$

$$\text{Volume of sphere} = \frac{4}{3}\pi r^3$$

$$\frac{4}{3}\pi 3^3 = \frac{4}{3}\pi 27 = 9 \times 4\pi = 36\pi$$

The maximum volume of the sphere that can be placed inside the cuboid is $36\pi \text{ cm}^3$.

Video Solution:**Q6 Text Solution:**

When a sphere is inscribed within a cube (or carved out with maximum volume from it), the diameter of the sphere will be equal to the side length of the cube.

Given, side length of the cube $a = 10$ cm.

Diameter of the largest inscribed sphere = $a = 10$ cm.

Radius r of the largest inscribed sphere = 2

$$\text{Diameter} = \frac{10}{2} = 5 \text{ cm.}$$

$$\text{Volume of sphere} = \frac{4}{3}\pi r^3$$

$$\text{Volume} = \frac{4}{3}\pi (5)^3$$

$$\text{Volume} = \frac{4}{3}\pi (125)$$

$$\text{Volume} = \frac{500\pi}{3} \text{ cm}^3$$

maximum volume of the sphere that can be carved out from the cube is $\frac{500\pi}{3} \text{ cm}^3$.

Video Solution:

Q7 Text Solution:

When a cube is inscribed in a sphere, all its vertices lie on the surface of the sphere. In this configuration, the longest diagonal of the cube (also known as the space diagonal) is equal to the diameter of the sphere.

Let the side length of the cube be a . The space diagonal of a cube is given by the formula $d = a\sqrt{3}$

radius of the sphere $R_{\text{sphere}} = 10$ cm.

The diameter of the sphere $= 2 \times R_{\text{sphere}} = 2 \times 10 = 20$ cm.

the space diagonal of the cube to the diameter of the sphere: $a\sqrt{3} = 20$ $a = \frac{20}{\sqrt{3}}$ cm

Volume of cube $= a^3$

$$\text{Volume} = \frac{(20)^3}{\sqrt{3}^3} = \frac{8000}{3\sqrt{3}} \text{ cm}^3$$

$$\text{Rationalizing} = \frac{8000}{3\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{8000\sqrt{3}}{9} \text{ cm}^3$$

Video Solution:**Q8 Text Solution:**

Height (H) = 18 cm

Base Radius (R) = 6 cm

The cut is made at 6 cm from the base. So, the height of the smaller cone (h) = Total Height - Height from base = $18 - 6 = 12$ cm.

the cut is parallel to the base, the smaller cone is similar to the original cone. The ratio of their heights is $k = \frac{h}{H} = \frac{12}{18} = \frac{2}{3}$

Due to similarity, the ratio of their radii is also k . So, the radius of the smaller cone (r) = $k \times R = \frac{2}{3} \times 6 = 4$ cm.

The ratio of the volumes of two similar cones is the cube of the ratio of their heights (or radii).

$$\frac{\text{Volume of small cone } V_{\text{small}}}{\text{Volume of big cone } V_{\text{big}}} = k^3 = \left(\frac{2}{3}\right)^3 = \frac{8}{27}$$

The volume of the frustum (V_{frustum}) is the volume of the big cone minus the volume of the small cone =

$$V_{\text{frustum}} = V_{\text{big}} - V_{\text{small}} = V_{\text{big}} - \frac{8}{27} V_{\text{big}} = \left(1 - \frac{8}{27}\right) V_{\text{big}} = \frac{19}{27} V_{\text{big}}$$

ratio of Smaller cone to frustum =

$$\frac{\frac{8}{27} V_{\text{big}}}{\frac{19}{27} V_{\text{big}}} = \frac{8}{19}$$

Video Solution:**Q9 Text Solution:**

Let V_s be the volume of the smaller cone and V_o be the volume of the original cone. Given $V_s = 1/64 V_o$.

For similar cones, the ratio of their volumes is the cube of the ratio of their heights. Let H_o be the original height and H_s be the height of the smaller cone.

$$\frac{V_s}{V_o} = \left(\frac{H_s}{H_o}\right)^3$$

$$\frac{1}{64} = \left(\frac{H_s}{H_o}\right)^3$$

$$\left(\frac{H_s}{H_o}\right) = \frac{1}{4}$$

$$H_s = \frac{1}{4} H_o$$

The height of the frustum is the difference between the original cone's height and the smaller cone's height.

Height of frustum = $H_o - H_s$

height of the frustum is 21 cm.

$$21 = H_o - \frac{1}{4} H_o$$

$$21 = \frac{3}{4} H_o$$

$$H_o = \frac{21 \times 4}{3} \Rightarrow H_o = 28 \text{ cm}$$

original height of the cone was 28 cm.



Video Solution:**Q10 Text Solution:**

Let $CSA_{original}$ be the curved surface area of the original cone and CSA_{small} be the curved surface area of the smaller cone formed at the top.

The curved surface area of the frustum is
 $CSA_{frustum} = CSA_{original} - CSA_{small}$.

$$CSA_{frustum} = \frac{7}{9} \times CSA_{original}.$$

$$\frac{7}{9} CSA_{original} = CSA_{original} - CSA_{small}$$

$$CSA_{small} = CSA_{original} - \frac{7}{9} \times CSA_{original}$$

$$CSA_{small} = \frac{2}{9} \times CSA_{original}$$

When a cone is cut parallel to its base, the smaller cone formed is similar to the original cone.

The ratio of their corresponding linear dimensions (heights, radii, slant heights) is k .

Ratio of their curved surface areas is k^2

$$k^2 = \frac{CSA_{small}}{CSA_{original}} = \frac{2}{9}$$

$$k = \sqrt{\frac{2}{9}} = \frac{\sqrt{2}}{3}$$

The ratio of the height of the smaller cone (h_{small}) to the original height ($H_{original}$) is k .

$$k = \frac{h_{small}}{H_{original}} \Rightarrow h_{small} = 4$$

$$\frac{\sqrt{2}}{3} = \frac{4}{H_{original}}$$

$$H_{original} = \frac{4 \times 3}{\sqrt{2}} = \frac{12}{\sqrt{2}} \Rightarrow \frac{12}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= 6\sqrt{2} \text{ cm}$$

Video Solution:**Q11 Text Solution:**

Let the radius of the top of the conical vessel be ' r '.

$$\text{So, } 2 \times \pi \times r = 352 \text{ m} \Rightarrow r = 56 \text{ m}$$

$$\text{So, volume} = \left(\frac{1}{3}\right) \pi r^2 h = \left(\frac{1}{3}\right) \times \left(\frac{22}{7}\right) \times 56^2 \times 60 = 197120 \text{ m}^3$$

$$\text{Now 75\% of this volume} = 147840 \text{ m}^3$$

$$\text{Hence, time taken} = 147840/60 = 2464$$

$$\text{seconds} = 41.06 \text{ minutes} = \mathbf{41} \text{ (Approximately)}$$

Video Solution:**Q12 Text Solution:**

For the outer cylinder, $r_2 = 9 \text{ cm}$, height $h = 40 \text{ cm}$

As thickness = 3 cm, thus for the inner cylinder

$r_1 = 6 \text{ cm}$, height $h = 40 \text{ cm}$

Hence, the total surface area of the hollow cylinder

= Lateral surface area of the inner cylinder +

Lateral surface area of the outer cylinder +

Area available due to thickness on top and bottom

$$\Rightarrow \text{TSA} =$$

$$2 \pi r_1 h + 2 \pi r_2 h + 2 \pi (r_2^2 - r_1^2)$$

$$\Rightarrow \text{TSA} =$$

$$2 \pi (9 \times 40 + 6 \times 40 + (9^2 - 6^2))$$

$$\Rightarrow \text{TSA} = \mathbf{1290\pi} \text{ (Ans)}$$

Video Solution:

**Q13 Text Solution:**

Let 'r' and 'h' be the radius and height of the solid cylinder respectively.

Given that, $r + h = 15$ and

$$2 \times \pi \times r(r + h) = 660$$

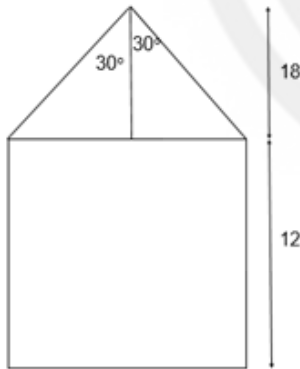
$$\Rightarrow 2 \times \pi \times r \times 5 = 660$$

$$\Rightarrow r = 7 \text{ cm}$$

$$\text{Thus, } h = 15 - 7 = 8 \text{ cm}$$

Hence, the volume of the cylinder =

$$\pi r^2 h = \frac{22}{7} \times 7 \times 7 \times 8 = 1232 \text{ cm}^3 \text{ (Ans)}$$

Video Solution:**Q14 Text Solution:**

Here the radius of the cylinder = radius of the cone = $6\sqrt{3}m$

As, the cone angle = 60° , thus, $\tan(30) = r/h$

$$\Rightarrow h = 6\sqrt{3} \times \sqrt{3} = 18 \text{ m}$$

Thus, volume of the building = Volume of the cone + Volume of the cylinder

$$\Rightarrow \text{Volume} = \pi r^2 h_1 + \frac{1}{3} \pi r^2 h_2$$

$\Rightarrow \text{Volume} =$

$$\pi \left(6\sqrt{3} \times 6\sqrt{3} \times 12 + \frac{1}{3} \times 6\sqrt{3} \times 6\sqrt{3} \times 18 \right)$$

$$\Rightarrow \text{Volume} = 1944\pi \text{ m}^3 \text{ (Ans)}$$

Video Solution:**Q15 Text Solution:**

Let the radius be 'r' and the height be 'h' for both the cone and cylinder.

$$\text{CSR of the cylinder} = 2\pi rh$$

$$\text{Slant height of the cone} = l = \sqrt{h^2 + r^2}$$

$$\text{So, CSA of cone} = \pi rl = \pi r \sqrt{h^2 + r^2}$$

$$\text{Now, } \frac{\pi r \sqrt{h^2 + r^2}}{2\pi rh} = 17 : 16$$

Solving this,

$$\Rightarrow r : h = 15 : 8 \text{ (Ans)}$$

Video Solution:**Q16 Text Solution:**

Area of a regular hexagon when the length of the smaller diagonal is given

$$= \frac{\sqrt{3}}{2} (\text{smaller diagonal})^2$$

$$= \frac{\sqrt{3}}{2} (14)^2$$

$$= \frac{\sqrt{3}}{2} \times 196$$

$$= 98\sqrt{3} \text{ cm}^2$$

Video Solution:

**Q17 Text Solution:**

On the basis of the above, the following equation can be formed

$$\frac{360}{x-2} + \frac{360}{x+3} = 12$$

$$\frac{1}{x-2} + \frac{1}{x+3} = \frac{1}{30}$$

$$\frac{x+3-x-2}{(x-2)(x+3)} = \frac{1}{30}$$

$$(x-2)(x+3) = 150$$

$$x^2 + x - 156 = 0$$

$$(x+13)(x-12) = 0$$

$$x = -13 \text{ and } x = 12$$

Since x cannot be negative, $x = 12$

Video Solution:**Q18 Text Solution:**

Let the side of the given regular hexagon be equal to 'x'

Then the inradius of the hexagon will be $\frac{\sqrt{3}x}{2}$

Also, the circumradius of the hexagon will be equal to the side of the hexagon, therefore it will be x

Thus, the required ratio of circumradius to inradius will be

$$x : \frac{\sqrt{3}x}{2}$$

$$= 2 : \sqrt{3}$$

Video Solution:**Q19 Text Solution:**

We know that each interior angle of a regular polygon having 'n' sides is given by

$$\frac{[(n-2)180]}{n}$$

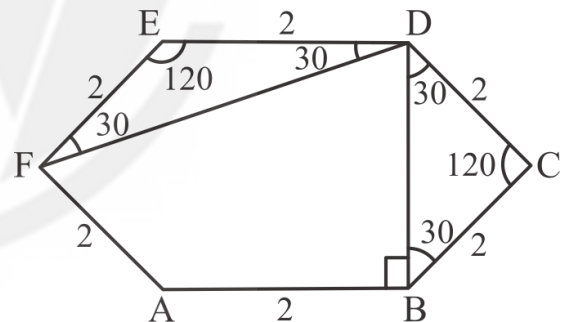
Therefore we have,

$$\frac{[(n-2)180]}{n} = 170$$

$$180n - 360 = 170n$$

$$10n = 360$$

$$n = 36.$$

Video Solution:**Q20 Text Solution:**

Given AD (longer diagonal) = 4 cm

Thus, side of the hexagon (a) = $\frac{4}{2} = 2$ cm

Area of the hexagon =

$$\frac{3\sqrt{3}}{2} \times a^2 = \frac{3\sqrt{3}}{2} \times 4 = 6\sqrt{3}$$

Area of Quadrilateral ABDF = Area of the hexagon - Area of triangle BCD - Area of triangle FDE

External angle of a hexagon = $\frac{360}{6} = 60$

Thus, Internal angle = $180 - 60 = 120$

Therefore, in triangle BCD, angle C is 120, so the remaining angles will be 30 each (Since BC



= CD being the sides of the hexagon)

Similarly, triangle FDE will also be a 120-30-30 triangle

Area of ABDF =

$$6\sqrt{3} - \frac{1}{2} \times 2 \times 2 \times \sin 120 - \frac{1}{2}$$

$$\times 2 \times 2 \times \sin 120$$

$$6\sqrt{3} - 4 \sin (180 - 60)$$

$$6\sqrt{3} - 4 \sin 60$$

$$6\sqrt{3} - 4 \times \frac{\sqrt{3}}{2}$$

$$6\sqrt{3} - 2\sqrt{3}$$

$$= 4\sqrt{3} \text{ cm}^2$$

Video Solution:



Q21 Text Solution:

Sum of interior angle in a regular polygon = $(n-2)180$

Sum of exterior angle in a regular polygon = 360

According to given condition, $(n-2)180 = 8(360)$

$$180n = 9 \times 360$$

$$n = 18$$

Number of sides = 18

$$\text{So number of diagonals} = {}^{18}C_2 - 18 = 153 - 18 =$$

$$135$$

Video Solution:



Q22 Text Solution:

Let the number of sides of polygons be $2a$ and $3a$

Interior angle of a regular polygon is

$\frac{n-2}{n} \times 180$ where 'n' denotes the number of sides of the polygon

We have,

$$\frac{\frac{2a-2}{2a} \times 180}{\frac{3a-2}{3a} \times 180} = \frac{3}{4}$$

$$= \frac{2a-2}{2a} \times \frac{3a}{3a-2} = \frac{3}{4}$$

Solving it, $a = 2$

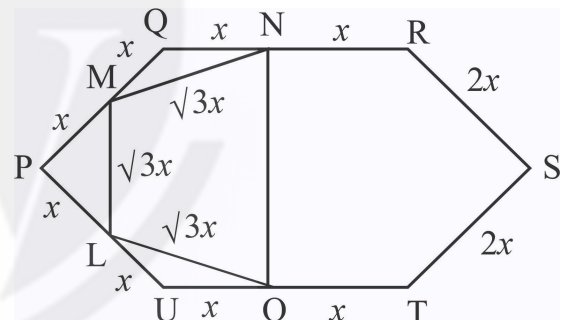
Hence the number of sides is 4 and 6 and the difference is 2.

Video Solution:



Q23 Text Solution:

Let the sides of the hexagon be of length $2x$ cm.



So area of the hexagon =

$$\frac{3\sqrt{3}}{2} (2x)^2 = 6\sqrt{3}x^2$$

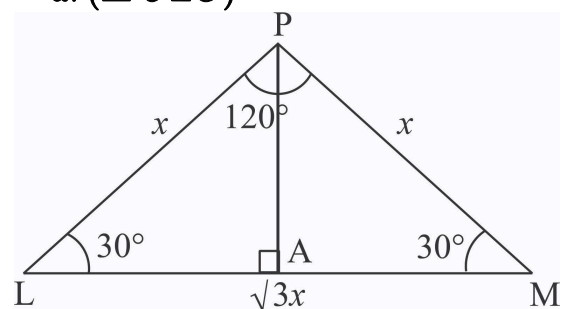
Area of half hexagon = PQNOU =

$$\frac{1}{2} \times 6\sqrt{3}x^2 = 3\sqrt{3}x^2$$

Area of quadrilateral MNOL = ar(PQNOU)

$$- \text{ar}(\triangle PML) - \text{ar}(\triangle QNM)$$

$$- \text{ar}(\triangle ULO)$$



Since all the interior angles of a regular hexagon are 120° , all the three triangles will form 30:30:120 triangles since the sides containing angle 120 are equal(x each). So all the three triangles will form 30:30:120 triangles with sides $x:x:\sqrt{3}x$.

Since all the three triangles are congruent(SSS), the area of all three will be equal.

If we draw PA perpendicular to LM, it will provide us with two 30:60:90 triangles with x opposing 90, so the side opposing 30, i.e., PA will be $\frac{x}{2}$, which will be the height of the triangle PML.

So area of triangle PLM =

$$\frac{1}{2} \times \frac{x}{2} \times \sqrt{3}x = \frac{\sqrt{3}}{4}x^2$$

So

Area of quadrilateral MNOL =

$\text{ar}(\text{PQNOU}) - 3(\text{ar}(\text{PLM}))$

$$= 3\sqrt{3}x^2 - \frac{3\sqrt{3}}{4}x^2$$

$$= \frac{9\sqrt{3}}{4}x^2$$

Since

$$6\sqrt{3}x^2 = 360$$

$$\Rightarrow x^2 = \frac{60}{\sqrt{3}} \text{ cm}^2$$

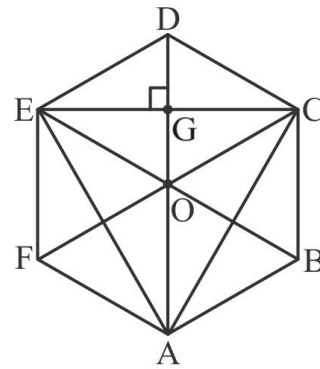
So $\text{ar}(\text{MNOL}) =$

$$\frac{9\sqrt{3}}{4}x^2 = \frac{9\sqrt{3}}{4} \times \frac{60}{\sqrt{3}} = 135 \text{ cm}^2$$

Video Solution:



Q24 Text Solution:



$$AD = 2OD = 2(OG + GD) = 2 \times 2 \times GD = 4GD$$

$$\frac{AD}{GD} = \frac{4}{1}$$

$$AD : GD = 4 : 1$$

$$\text{So, } AF : GD = 3 : 1$$

Now, $AG = 3 \text{ cm}$ (given)

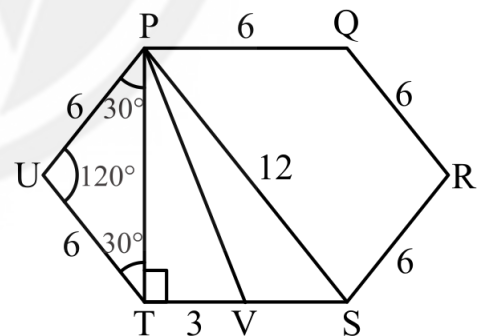
So, $GD = 1 \text{ cm}$

Therefore, $OD = 2GD = 2 \text{ cm} =$ (side length of the hexagon)

Video Solution:



Q25 Text Solution:



So all the sides are 6cm

If we join PT, it will form a 30:120:30 triangle

PUT, so $\angle UTP = 30^\circ$

Also since the sides of 30:120:30 triangle are in

$1:\sqrt{3}:1$ ratio,

$$PT = 6\sqrt{3} \text{ cm}$$

Also $\angle PTV$ will be 90°

So

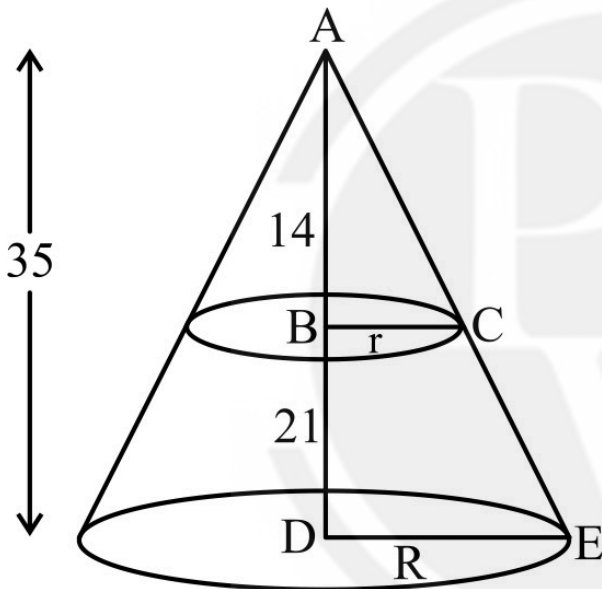


$$\begin{aligned}
 PV^2 &= PT^2 + TV^2 \\
 \Rightarrow PV^2 &= (6\sqrt{3})^2 + 3^2 \\
 \Rightarrow PV^2 &= 117 \\
 \Rightarrow PV &= \sqrt{117} \text{ cm}
 \end{aligned}$$

Video Solution:



Q26 Text Solution:



Lets consider the radius of the original cone be R and that of the resultant cone be r .

Also the height of the resultant cone = $35 - 21 = 14$ cm

If a smaller cone is cut out from larger cone then the ratio of the radius of the two cones is same as the ratio of their respective heights, so

$$\frac{r}{R} = \frac{14}{35} = \frac{2}{5}$$

Volume of larger cone = $\frac{1}{3}\pi R^2 (35)$

Volume of smaller cone =

$$\frac{1}{3}\pi r^2 (14) = \frac{1}{3}\pi \left(\frac{2}{5}R\right)^2 (14) = \frac{56}{75}\pi R^2$$

Volume of frustum of cone =

$$\begin{aligned}
 &\frac{1}{3}\pi R^2 (35) - \frac{56}{75}\pi R^2 = \frac{1}{3}\pi R^2 \left(35 - \frac{56}{25}\right) \\
 &= \frac{273}{25}\pi R^2
 \end{aligned}$$

So

Volume of frustum of cone- Volume of smaller cone = 7194

$$\frac{273}{25}\pi R^2 - \frac{56}{75}\pi R^2 = 7194$$

$$\Rightarrow \frac{819-56}{75}\pi R^2 = 7194$$

$$\Rightarrow R^2 = \frac{7194 \times 7 \times 75}{763 \times 22}$$

$$\Rightarrow R^2 = 225$$

$$\Rightarrow R = 15 \text{ cm} \quad \left(\text{We should ignore } -15 \text{ as a physical unit can't be negative}\right)$$

So volume of the frustum of cone =

$$\frac{273}{25}\pi R^2 = \frac{273}{25} \times \frac{22}{7} \times 225 = 7722 \text{ cm}^3$$

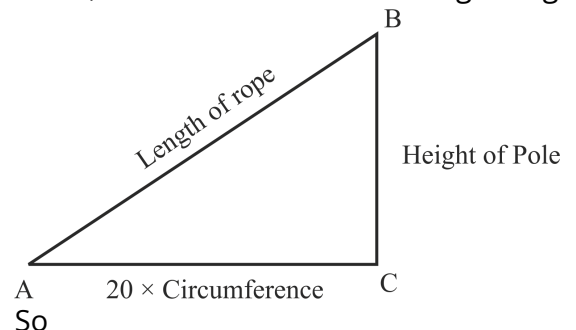
Video Solution:



Q27 Text Solution:

If a cylinder is opened, it forms a rectangle with height as one of the dimensions and circumference as the other dimension.

Since the rope is wrapped around the pole 20 times, we will obtain the following triangle



So

Circumference of the pole =

$$2\pi r = 2 \times \frac{22}{7} \times \frac{7}{2} = 22 \text{ cm}$$

So

$$AC = 20 \times 22 = 440 \text{ cm} = 4.4 \text{ m}$$



$$BC = 24 \text{ m}$$

So

$$\text{Length of rope} = AB =$$

$$\begin{aligned}\sqrt{AC^2 + BC^2} &= \sqrt{(4.4)^2 + (24)^2} \\ &= \sqrt{4^2 [(1.1)^2 + (6)^2]} = 4\sqrt{1.21 + 36} \\ &= 4\sqrt{37.21} = 4 \times 6.1 = 24.4 \text{ m} \\ &= 2440 \text{ cm}\end{aligned}$$

Video Solution:



Q28 Text Solution:

Volume of the tank (pyramid)=

$$\begin{aligned}\frac{1}{3} \times \text{Area of the base} \times \text{height} \\ = \frac{1}{3} \times \frac{3\sqrt{3}}{2} \times (17.5)^2 \times (44\sqrt{3})\end{aligned}$$

$$= 3 \times 17.5 \times 17.5 \times 22 \text{ cm}^3$$

Rate of flow of water through the pipe per second = Volume of cylinder of water formed by flow in one second

$$\begin{aligned}&= \pi \times \left(\frac{10.5}{2 \times 10 \times 10}\right)^2 \times \left(\frac{400 \times 100}{3600}\right) \\ &= \frac{22 \times 10.5 \times 10.5 \times 400 \times 100}{7 \times 2 \times 2 \times 10 \times 10 \times 3600} \\ &= \frac{22 \times 10.5 \times 10.5}{7 \times 36} \text{ cm}^3\end{aligned}$$

So time required =

$$\begin{aligned}&\frac{\text{Volume of the tank}}{\text{Rate of flow of water through the pipe per second}} \\ &= \frac{3 \times 17.5 \times 17.5 \times 22}{\frac{22 \times 10.5 \times 10.5}{7 \times 36}}\end{aligned}$$

(Changing everything into cm)

$$\begin{aligned}&= \frac{3 \times 17.5 \times 17.5 \times 22 \times 7 \times 36}{22 \times 10.5 \times 10.5} \\ &= \frac{3 \times 5 \times 5 \times 7 \times 36}{3 \times 3} \\ &= 2100 \text{ seconds}\end{aligned}$$

Video Solution:



Q29 Text Solution:

The square face of the cuboid has an area of 64 sq. cm so the sides of that square face will be = 8 cm

Also since the cube has a TSA of 384 sq cm, so

$$\text{side of cube} = \sqrt{\frac{384}{6}} = 8 \text{ cm}$$

Since there is no wastage of material and each cube has been casted from a single metal, this means that all the cuboids will have length exactly divisible by the side length of the cubes, i.e., 8 cm, only option c fits the criteria.

Video Solution:



Q30 Text Solution:

A cuboid of maximum possible volume inside a sphere is a cube. So longest diagonal of the cube is equal to the diameter of the sphere.

If x is the side of the cube, then the longest

diagonal is $\sqrt{3}x$.

So

$$\sqrt{3}x = 14\sqrt{3}$$

$$\Rightarrow x = 14 \text{ cm}$$

So volume of the remaning sphere = Volume of sphere - Volume of cube

$$\begin{aligned}&= \frac{4}{3}\pi r^3 - x^3 \\ &= \frac{4}{3} \times \frac{22}{7} \times (7\sqrt{3})^3 - 14^3 \\ &= 392 \times 11\sqrt{3} - 392 \times 7 \\ &= 392(11\sqrt{3} - 7) \text{ cm}^3\end{aligned}$$



Video Solution:



[Android App](#)



[iOS App](#)



[PW Website](#)

